

Week 8: Options Market Making & Dynamic Hedging

Avellaneda-Stoikov with a vector-valued (Greek) inventory

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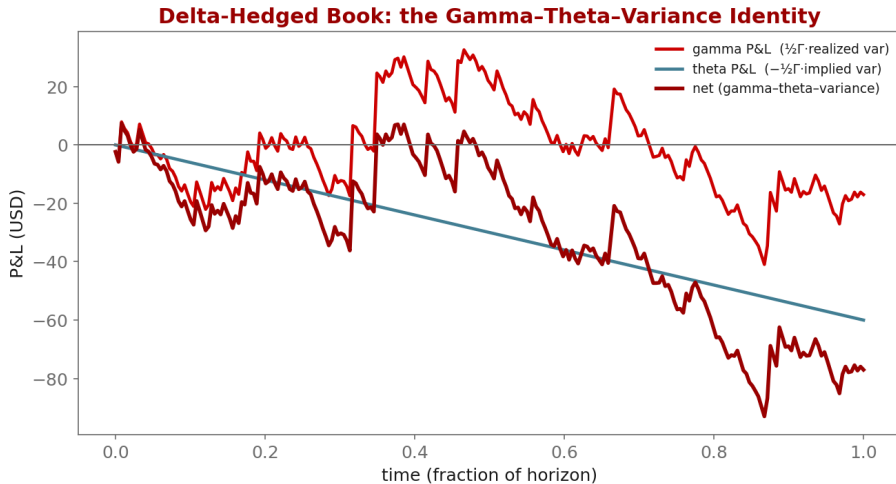
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1. Learning objectives
2. Lecture 1: Multi-Greek Inventory & HJB
3. Lecture 2: Dynamic Hedging
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: week8_hedge.py
8. Summary & key takeaways
9. Glossary of key terms
10. Reading
11. References

- ▶ Set up the El Aoud-Abergel multi-Greek framework
- ▶ Understand the quadratic inventory penalty
- ▶ Quantify the P&L of a delta-hedged book

- ▶ Inventory becomes a vector $q = (\text{Delta}, \text{Gamma}, \text{vega}, \dots)$
- ▶ Quadratic inventory penalty; reduces to AS in 1-D
- ▶ Multi-Greek HJB and the separation ansatz

- ▶ Continuous vs discrete delta hedging
- ▶ The gamma-theta-variance identity
- ▶ Perp futures as the hedge instrument; funding cost



$$\text{penalty} = \frac{1}{2} q^T \Sigma q$$

Why: Inventory risk for an options book is multi-dimensional (delta, gamma, vega...). A quadratic form is the local (mean-variance) risk of the Greek vector; Σ is calibrated to the empirical covariance of the Greeks, so cross-risks are priced, not just diagonal ones. It reduces to AS's q^2 penalty when there is one Greek.

$$d\Pi = \frac{1}{2} \Gamma S^2 (\sigma_{\text{real}}^2 - \sigma_{\text{impl}}^2) dt$$

Why: The gamma-theta-variance identity: a delta-hedged option earns (or pays) the gap between realized and implied variance, scaled by dollar gamma. This single equation explains 80

- ▶ Greeks per contract (for the inventory vector and Sigma):
 - ▶ GET /open/option/detail/v1/{symbol} -> delta, gamma, vega, theta, rho
- ▶ Hedge instrument + carry:
 - ▶ Perp book: GET /open/futures/order/orderbook/v1/{symbol}
 - ▶ Funding: GET /open/public/fundingRate/v1?symbol=BTCUSD
- ▶ Offline: `coincall_funding_rates_2025Q4.parquet` for hedge-cost accounting

Cross-check exact paths at <https://docs.coincall.com/>

```
import torch
def book_greeks(positions, greeks):
    # positions: (n,), greeks: (n,5) columns [delta,gamma,vega,theta,rho]
    return (positions[:, None] * greeks).sum(0) # aggregate inventory vector

def delta_hedge(net_delta, perp_pos, threshold=0.1):
    if net_delta.abs() > threshold: # rebalance only when breached
        trade = -net_delta # hedge with perp futures
        return perp_pos + trade, trade
    return perp_pos, torch.tensor(0.0)
```

- ▶ Options inventory is a VECTOR of Greeks, penalized by $(1/2) q^T \Sigma q$
- ▶ The multi-Greek HJB reduces to Avellaneda-Stoikov in the 1-D limit
- ▶ Gamma-theta-variance identity explains $\sim 80\%$ of a hedged book's P&L
- ▶ Hedge delta with the perpetual; account for the funding rate
- ▶ Calibrate Sigma to the empirical Greek covariance, not the identity

Greek inventory vector aggregated (delta, gamma, vega, ...) exposure of the book

Quadratic penalty the risk term $(1/2) \mathbf{q}^T \Sigma \mathbf{q}$ on the Greek vector

Delta hedging neutralizing directional risk using the underlying or a perpetual

Gamma scalping P&L earned by re-hedging a long-gamma position

Gamma-theta-variance identity hedged P&L = $(1/2) \Gamma S^2 (\text{realized var} - \text{implied var}) dt$

Funding rate periodic payment between perpetual-swap longs and shorts

El Aoud-Abergel the multi-Greek option market-making framework

- ▶ El Aoud-Abergel (2015); Taleb (1997); Cartea et al., Ch. 11

- ▶ El Aoud, S., & Abergel, F. (2015). A stochastic control approach to option market making. *Market Microstructure and Liquidity*, 1(1), 1550006.
- ▶ Taleb, N. N. (1997). *Dynamic Hedging: Managing Vanilla and Exotic Options*. Wiley.
- ▶ Cartea, Jaikumar, & Penalva (2015). *Algorithmic and High-Frequency Trading*, Ch. 11.